BLOCKCHAIN TECHNOLOGY







INDEX

Sr. No.	Topic
1.	What is Blockchain Technology?
2.	Fundamentals of Blockchain
3.	Types of Blockchain Technology
4.	Why is Blockchain important?
5.	How does Blockchain Work?
6.	Transaction process in Blockchain
7.	Decentralization in Blockchain
8.	Why is decentralization important?
9.	Applications of Blockchain
10.	History of Blockchain Technology
11.	Bitcoin vs. Blockchain
12.	Advantages and Disadvantages of Blockchain Technology
13.	Conclusion
14.	References

What is Blockchain Technology?

A blockchain is a distributed database or ledger that is shared among the nodes of a computer network. As a database, a blockchain stores information electronically in digital format. Blockchains are best known for their crucial role in cryptocurrency systems, such as Bitcoin, for maintaining a secure and decentralized record of transactions. The innovation with a blockchain is that it guarantees the fidelity and security of a record of data and generates trust without the need for a trusted third party.



Fundamentals of Blockchain

1. Public Distributed Ledgers

• A blockchain is a decentralized public distributed ledger that is used to record transactions across many computers.

• A distributed ledger is a database that is shared among the users of the blockchain network.

• The transactions are accessed and verified by users associated with the bitcoin network, thereby making it less prone to cyberattack.

2. Encryption

• Blockchain eliminates unauthorized access by using the cryptographic algorithm (SHA256) to ensure the blocks are kept secure.

• Each user in the blockchain has their key.

3. Proof of Work

Proof of work (PoW) is a method to validate transactions in a blockchain network by solving a complex mathematical puzzle called mining.

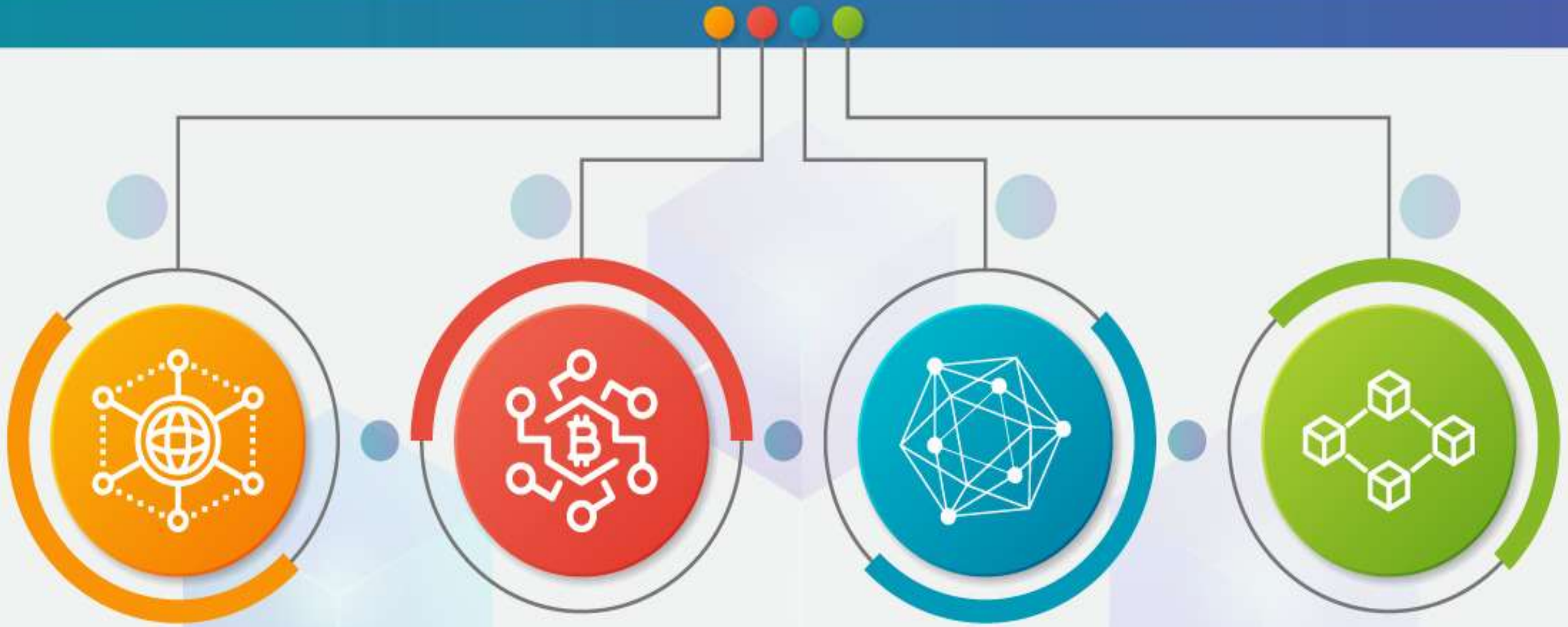
Note: Users trying to solve the puzzle are called miners.

4. Mining

In Blockchain, when miners use their resources (time, money, electricity, etc.) to validate a new transaction and record them on the public ledger, they are given a reward.

Note: As a reward, the miner gets 12.5 BTC (bitcoins)

Four Types Of Blockchain Technology



PUBLIC BLOCKCHAIN

PRIVATE BLOCKCHAIN

CONSORTIUM BLOCKCHAIN

HYBRID BLOCKCHAIN

1. Public Blockchain

These blockchains are completely open to following the idea of decentralization. They don't have any restrictions, anyone having a computer and internet can participate in the network.

2. Private Blockchain

These blockchains are not as decentralized as the public blockchain only selected nodes can participate in the process, making it more secure than the others.

3. Consortium Blockchain

It is a creative approach that solves the needs of the organization. This blockchain validates the transaction and also initiates or receives transactions. It is also known as Federated Blockchain.

4. Hybrid Blockchain

It is the mixed content of the private and public blockchain, where some part is controlled by some organization and other makes are made visible as a public blockchain.

Why is Blockchain important?

Blockchain is an especially promising and revolutionary technology because it helps reduce security risks, stamp out fraud and bring transparency in a scalable way.
Originally popularized in the 2010s by its association with cryptocurrency and NFTs, blockchain technology has since evolved to become a management solution for all types of global industries. Today, you can find blockchain technology providing transparency for the food supply chain, securing healthcare data, innovating gaming and overall changing how we handle data and ownership on a large scale.

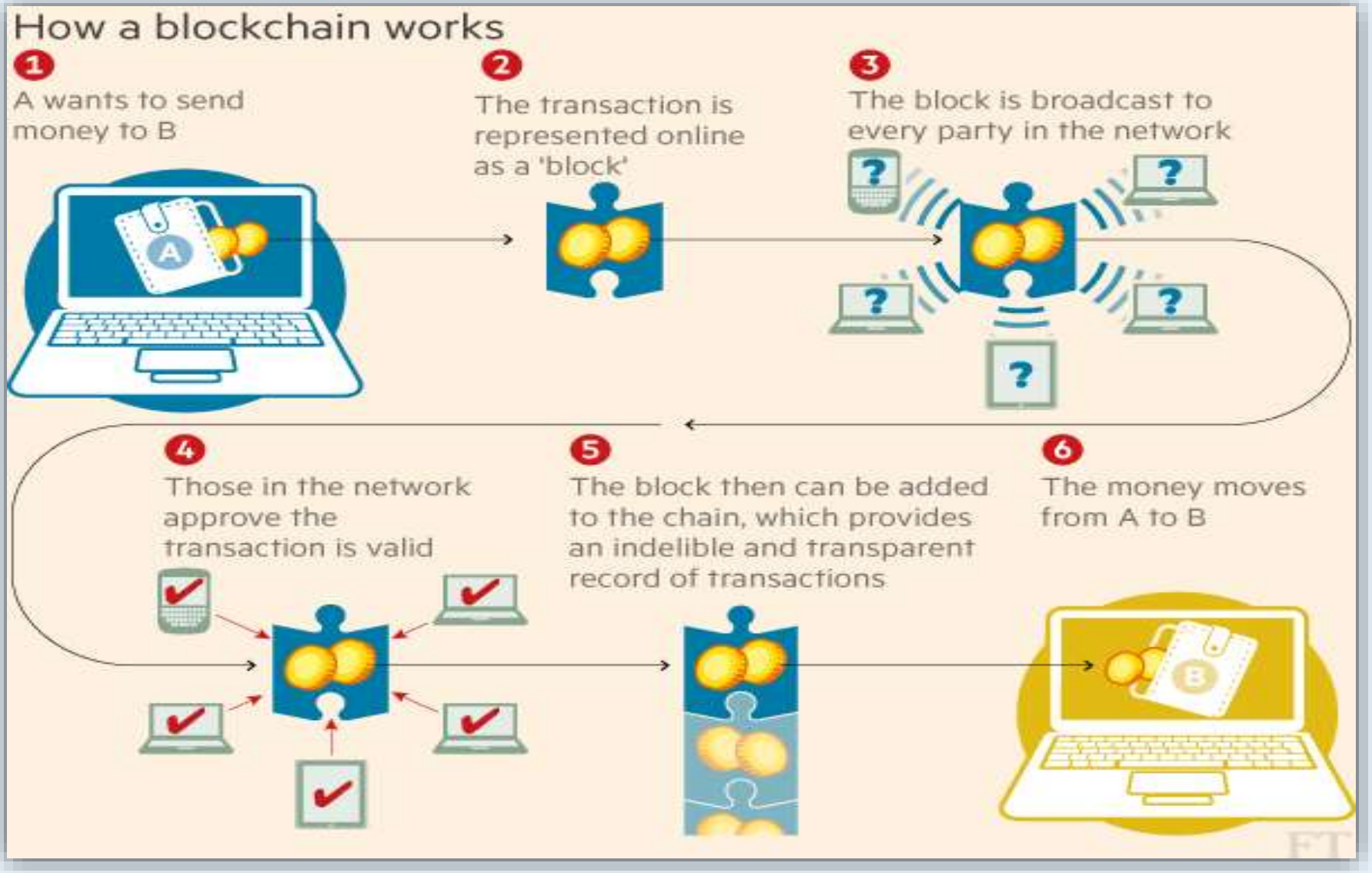


How Does Blockchain Work?

The goal of blockchain is to allow digital information to be recorded and distributed, but not edited. In this way, a blockchain is the foundation for immutable ledgers, or records of transactions that cannot be altered, deleted, or destroyed. This is why blockchains are also known as a distributed ledger technology (DLT).

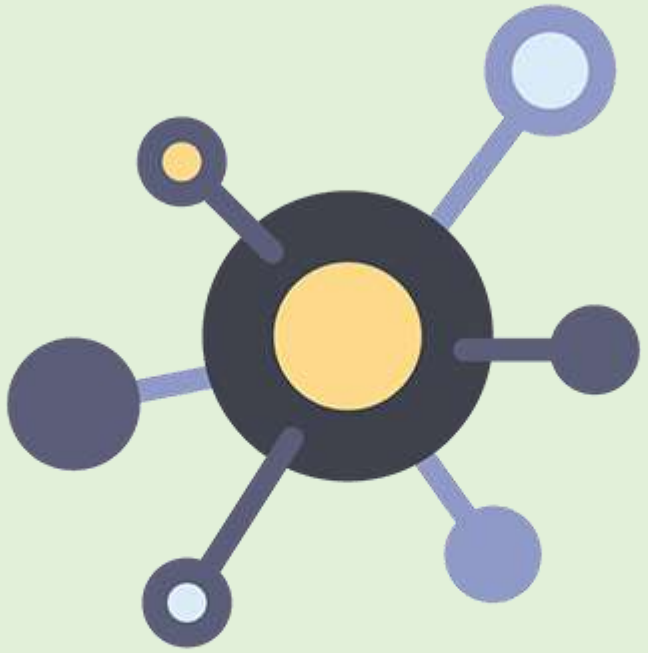
First proposed as a research project in 1991, the blockchain concept predated its first widespread application in use: Bitcoin, in 2009. In the years since, the use of blockchains has exploded via the creation of various cryptocurrencies, decentralized finance (DeFi) applications, non-fungible tokens (NFTs), and smart contracts.

Transaction Process in Blockchain



Decentralization in Blockchain

In the blockchain, decentralization alludes to the transfer of supervision and decision-making from a centralized association (individual, corporation, or group of people) to a dispersed network. Decentralized networks endeavor to decrease the degree of trust that members should put in each other and dissuade their capacity to put forth authority or command over each other in manners that corrupt the potency of the network.



DECENTRALIZED

Why is decentralization important?

Decentralization is actually not a new idea. When assembling a technological arrangement, three essential network structures are commonly thought of: centralized, decentralized, and disseminated. While blockchain technologies frequently utilize decentralized networks, a blockchain application itself simply can’t be sorted as being decentralized or not. Maybe, decentralization is a sliding scale and should be applied to all parts of a blockchain application. By decentralizing the administration of and admittance to assets in an application, more prominent and more pleasant assistance can be accomplished. Decentralization typically has a few setoffs, for example, lower exchange. However, such setoffs are worth it in comparison to the improved security and services they provide.

Applications Of Blockchain



Digital IDs



Bitcoin



Real estate



Voting



Payment and Transfers



Heath Care

Internet of Things





Online music



Banking



Law Enforcement

1. Money transfers

The original concept behind the invention of blockchain technology is still a great application. Money transfers using blockchain can be less expensive and faster than using existing money transfer services.

2. Financial exchanges

Many companies have popped up over the past few years offering decentralized cryptocurrency exchanges. Using blockchain for exchanges allows for faster and less expensive transactions.

3. Lending

Lenders can use blockchain to execute collateralized loans through smart contracts. Smart contracts built on the blockchain allow certain events to automatically trigger things like a service payment, a margin call.

4. Insurance

Using smart contracts on a blockchain can provide greater transparency for customers and insurance providers. Recording all claims on a blockchain would keep customers from making duplicate claims for the same event.

5. Secure Personal Information

Blockchain technology can be used to secure access to identifying information while improving access for those who need it in industries such as travel, healthcare, finance, and education..

6. Voting

If personal identity information is held on a blockchain, from also being able to vote using blockchain technology. Using blockchain technology can make sure that nobody votes twice, only eligible voters are able to vote.

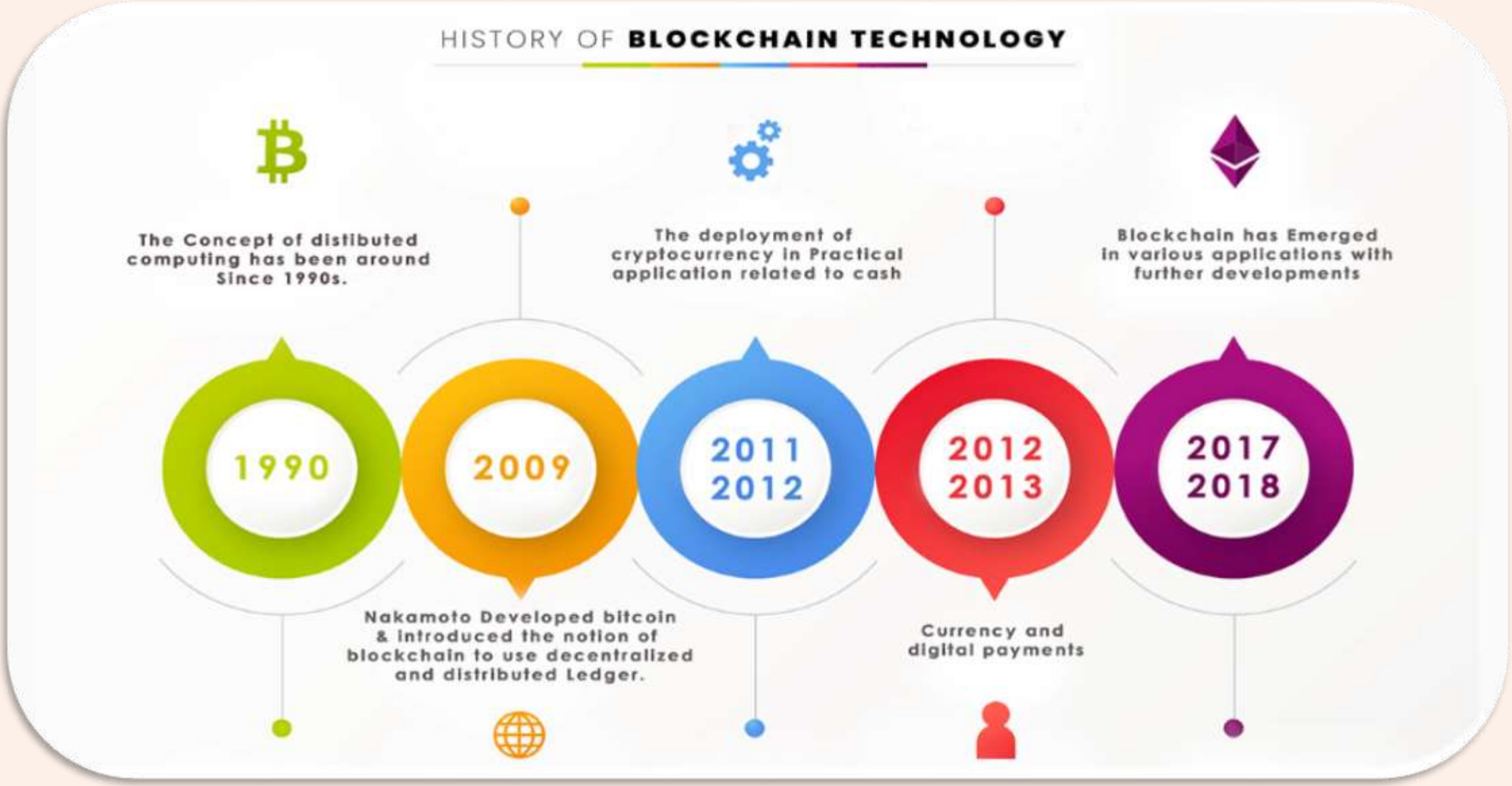
7. Real Estate

Using blockchain technology to record real estate transactions can provide a more secure and accessible means of verifying and transferring ownership. That can speed up transactions, reduce paperwork, and save money.

8. Health Care

Keeping medical records on a blockchain can allow doctors and medical professionals to obtain accurate and up-to-date information on their patients. That can ensure that patients seeing multiple doctors get the best care possible.

History of Blockchain Technology



Bitcoin vs. Blockchain

Blockchain technology was first outlined in 1991 by Stuart Haber and W. Scott Stornetta, two researchers who wanted to implement a system where document timestamps could not be tampered with. But it wasn't until almost two decades later, with the launch of Bitcoin in January 2009, that blockchain had its first real-world application.

The Bitcoin protocol is built on a blockchain. In a research paper introducing the digital currency, Bitcoin's pseudonymous creator, Satoshi Nakamoto, referred to it as "a new electronic cash system that's fully peer-to-peer, with no trusted third party."



BLOCKCHAIN

Advantages of Blockchain

1.**Open:** One of the major advantages of blockchain technology is that it is accessible to all means anyone can become a participant in the contribution to blockchain technology.

2.**Verifiable:** Blockchain technology is used to store information in a decentralized manner so everyone can verify the correctness of the information by using zero-knowledge.

3.**Permanent:** Records or information which is stored using blockchain technology is permanent means one needs not worry about losing the data.

4.**Tighter Security:** Blockchain uses hashing techniques to store each transaction on a block that is connected to each other so it has tighter security.

5.**Efficiency**: Blockchain removes any third-party intervention between transactions and removes the mistake making the system efficient and faster. Settlement is made easier and smooth.

6.**Cost Reduction**: As blockchain needs no third man it reduces the cost for the businesses and gives trust to the other partner.

Disadvantages of Blockchain Technology

1.Scalability: It is one of the biggest drawbacks of blockchain technology as it cannot be scaled due to the fixed size of the block for storing information.

2.Energy Consuming: For verifying any transaction a lot of energy is used so it becomes a problem according to the survey it is considered that 0.3 percent of the world's electricity had been used by 2018 in the verification of transactions done using blockchain technology.

3.Time-Consuming: To add the next block in the chain miners need to compute nonce values many times so this is a time-consuming process and needs to be speed up to be used for industrial purposes.

4.Legal Formalities: In some countries, the use of blockchain technology applications is banned like cryptocurrency due to some environmental issues.

5.Storage: Blockchain databases are stored on all the nodes of the network creates an issue with the storage, increasing number of transactions will require more storage.

Conclusion

The Bitcoin is the first successful implementation of blockchain. Today, the world has found applications of blockchain technology in several industries, where the trust without the involvement of a centralized authority is desired. So welcome to the world of Blockchain.

Blockchain technology is only going to grow in the fields of business, finance, law, medicine, and real estate. Whether you're an experienced Blockchain developer, or you're aspiring to break into this exciting industry, enrolling in our Blockchain Certification Training program will help individuals with all levels of experience to learn Blockchain developer techniques and strategies.

References

1. https://en.wikipedia.org/wiki/Blockchain
2. https://www.investopedia.com/terms/b/blockchain.asp
3. https://www.pngegg.com/en/search?q=blockchain+Technology
4. https://www.simplilearn.com/tutorials/blockchain-tutorial/why-is-blockchain-important
5. https://www.geeksforgeeks.org/advantages-and-disadvantages-of-blockchain/
6. https://builtin.com/blockchain



Made By : Mr. Sahil Bansal